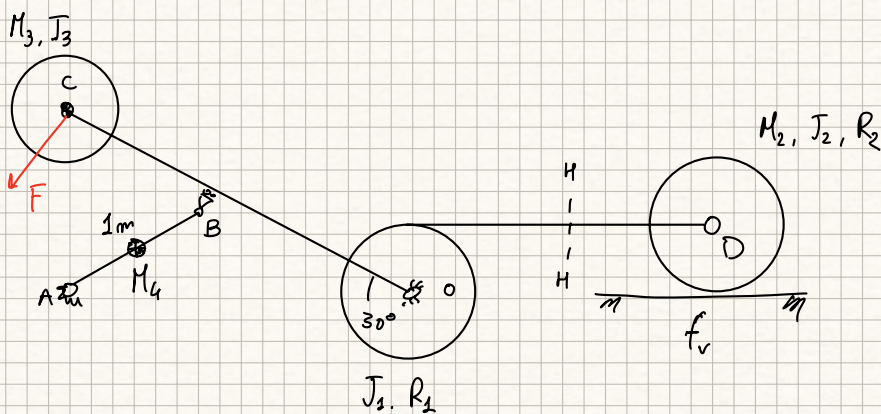
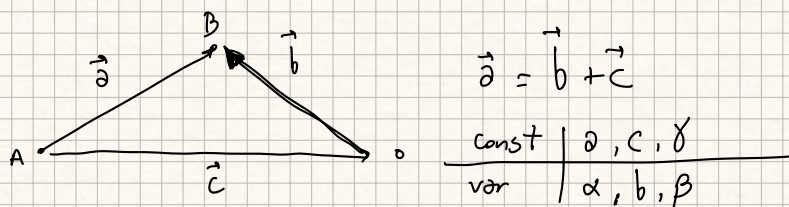




(1)



Posizioni

$$\begin{cases} a \cos \alpha = b \cos \beta + c \cos \gamma \\ a \sin \alpha = b \sin \beta + c \sin \gamma \end{cases}$$

$$\begin{aligned} a &= 1 \text{ m} & b &= 1.5 \text{ m} \\ \alpha &= 45^\circ & \beta &= 150^\circ \end{aligned}$$

Velocità

$$\begin{cases} -a \dot{\alpha} \sin \alpha = \dot{b} \cos \beta - b \dot{\beta} \sin \beta \\ a \dot{\alpha} \cos \alpha = \dot{b} \sin \beta + b \dot{\beta} \cos \beta \end{cases}$$

$$\Rightarrow \dot{b} = -0.9659 \text{ m/s}, \quad \dot{\beta} = 0.1725 \text{ rad/s}$$

Accelerazioni:

$$\begin{cases} -a \ddot{\alpha} \sin \alpha - a \dot{\alpha}^2 \cos \alpha = \ddot{b} \cos \beta - 2\dot{b} \dot{\beta} \sin \beta - b \ddot{\beta} \sin \beta - b \dot{\beta}^2 \cos \beta \\ a \ddot{\alpha} \cos \alpha - a \dot{\alpha}^2 \sin \alpha = \ddot{b} \sin \beta + 2\dot{b} \dot{\beta} \cos \beta + b \ddot{\beta} \cos \beta - b \dot{\beta}^2 \sin \beta \end{cases}$$

$$\Rightarrow \ddot{b} = -0.6624 \text{ m/s}^2, \quad \ddot{\beta} = 1.0387 \text{ rad/s}^2$$

$$\vec{u}_D = -|\dot{\beta}| R_1 \vec{i} = (-0.0863 \vec{i}) \text{ m/s} \Rightarrow \vec{\omega}_D = \frac{|\vec{u}_D|}{R_2} \vec{k} = (0.0719 \vec{k}) \text{ rad/s}$$

$$\vec{a}_D = -|\ddot{\beta}| R_1 \vec{i} = (-0.5194 \vec{i}) \text{ m/s}^2 \Rightarrow \vec{\dot{\omega}}_D = (0.4328 \vec{k}) \text{ rad/s}^2$$

$$\textcircled{2} \quad \frac{dE_c}{dt} = J_1 \dot{\beta} \ddot{\beta} + J_3 \dot{\beta} \ddot{\beta} + M_3 v_c a_c + M_4 v_G a_G + J_4 \dot{\alpha} \ddot{\alpha} + J_2 \omega_D \dot{\omega}_D + M_2 v_D a_D$$

$$|v_c| = \dot{\beta} \cdot \overline{OC} = 0.366 \text{ m/s} \quad |v_a| = 0.5 \text{ m/s}$$

$$|a_c| = \dot{\beta} \cdot \overline{OC} = 2.08 \text{ m/s}^2 \quad |a_a| = 0.5 \text{ m/s}^2$$

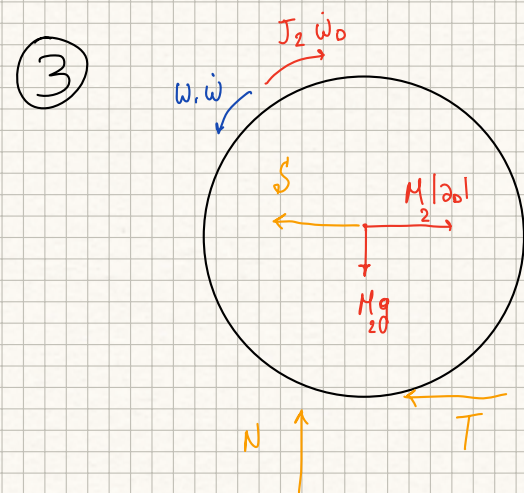
$$\frac{dE_c}{dt} = \underbrace{J_1 \dot{\beta} \ddot{\beta}}_{0.036} + \underbrace{J_3 \dot{\beta} \ddot{\beta}}_{0.216} + \underbrace{M_3 v_c a_c}_{1.43} + \underbrace{M_4 v_G a_G}_1 + \underbrace{J_2 \omega_D \dot{\omega}_D}_{0.0374} + \underbrace{M_2 v_D a_D}_{0.225}$$

$$\frac{dE_c}{dt} = 2.94 \text{ W}$$

$$\Sigma W_{ATT} = F |v_c| + \underbrace{M_a |g| v_a \cos 45}_{13.87} + \underbrace{M_3 |g| v_c \cos 30}_{5.88} - \underbrace{f_v R_2 M_2 g \omega_D}_{-0.212}$$

$$F |v_c| + 19.54 = 2.94$$

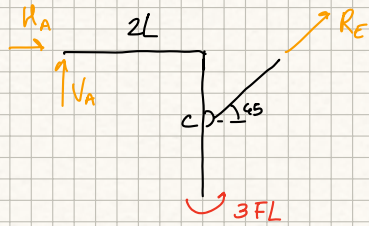
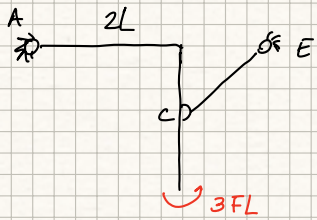
$$\Rightarrow F = -47.97 \text{ N}$$



$$\left. \begin{array}{l} N = M_2 g \\ \Sigma R_2 - J_2 \dot{\omega}_D - f_v R_2 M_2 g - M_2 |a_D| R_2 = 0 \end{array} \right\}$$

$$s = \underbrace{\frac{J_2 \dot{\omega}_D}{R_2}}_{0.634} + \underbrace{f_v M_2 g}_{2.45} + \underbrace{M_2 |a_D|}_{2.6} = 5.48 \text{ N}$$

Ex 2

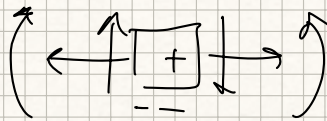
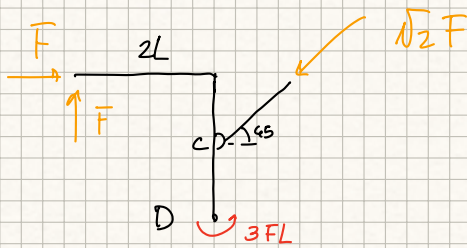


$$\sum M_A = 0 \Rightarrow R_E \frac{\sqrt{2}}{2} \cdot 3L + 3FL = 0$$

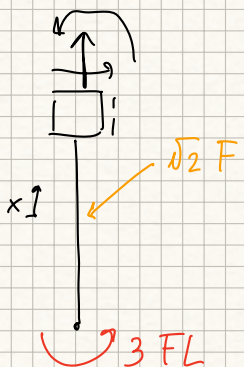
$$\Rightarrow R_E = -\frac{2F}{\sqrt{2}} = -\sqrt{2}F$$

$$H_A + R_E \frac{\sqrt{2}}{2} = 0 \Rightarrow H_A = F$$

$$V_A + R_E \frac{\sqrt{2}}{2} = 0 \Rightarrow V_A = F$$



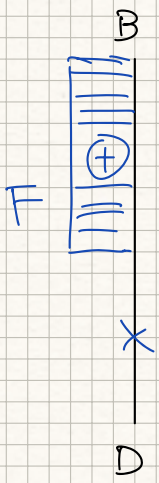
$$\begin{cases} N = 0 \\ T = 0 \\ M_{cl} = -3FL \end{cases}$$



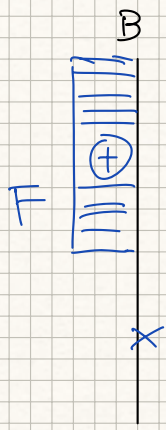
$$\begin{cases} N = F \\ T = F \\ M + 3FL - Fx = 0 \end{cases}$$

$$M(0) = -3FL$$

$$M(L) = -2FL$$



$$\leftarrow \boxed{N} \rightarrow$$



$$\uparrow \boxed{T} \downarrow$$



$$\left(\boxed{M_e} \right)$$

ES 3

$$W_1 = C_m \dot{W}_m - J_m \ddot{W}_m$$

$$\frac{dE}{dt}^{(0)} = 4 J \tau^2 \dot{W}_m \ddot{W}_m + M R^2 \tau^2 \dot{W}_m \ddot{W}_m = J_u^* \dot{W}_m \ddot{W}_m = 5.72 \dot{W}_m \ddot{W}_m$$

$$\begin{aligned} \Sigma W_{ATT} &= M g \sin \alpha \tau R \dot{W}_m - f_v R |\dot{W}_u 2 N_p| - f_v R |\dot{W}_u 2 N_A| = \\ &= M g \sin \alpha \tau R \dot{W}_m - f_v R \tau \dot{W}_m |2 N_p + 2 N_A| = M g \tau R (\sin \alpha - f_v \cos \alpha) \dot{W}_m \\ &= C_u^* \dot{W}_m = 32.97 \dot{W}_m \end{aligned}$$

1) A REGIME

$$W_2 = C_u^* \dot{W}_m > 0 \Rightarrow \text{MOTO RETROGRADO}$$

$$C_m \dot{W}_m + \eta_R C_u^* \dot{W}_m = 0 \Rightarrow C_m = -23.08 \text{ Nm}$$

2) ALLO SPUNTO : $W_2 = C_u^* \dot{W}_m - J_u^* \ddot{W}_m$

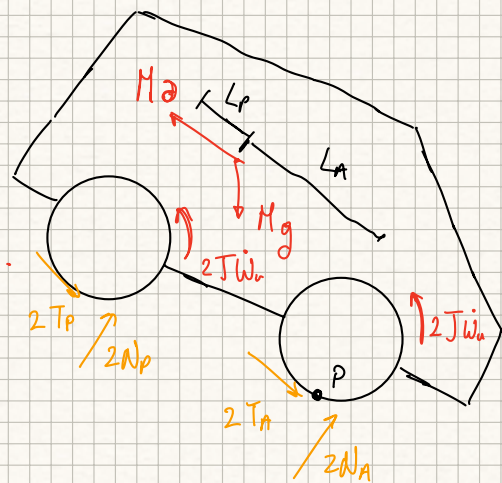
ASSUMO MOTO DIRETTO

$$\eta_D C_{mo} - \eta_D J_m \ddot{W}_m + C_u^* - J_u^* \ddot{W}_m = 0$$

$$\dot{W}_m = \frac{\eta_D C_{mo} + C_u^*}{\eta_D J_m + J_u^*} = 32.82 \text{ rad/s}^2, \quad W_1 = (C_{mo} - J_m \ddot{W}_m) \dot{W}_m \approx 193 \dot{W}_m > 0 \quad \text{OK}$$

$$a = \tau \ddot{W}_m R = 1.96 \text{ m/s}^2$$

3) Aderenza ruote posteriori da pt 2)



$$2N_p + 2N_A = Mg \cos \alpha$$

$$\Sigma M_p = 0 \Rightarrow + 4J \ddot{\omega}_m \tau + M \tau \ddot{\omega}_m R H + 2N_A f_v R +$$

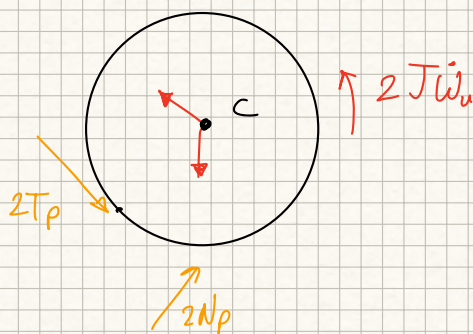
$$- 2N_p (L_p + L_A - f_v R) + Mg \cos \alpha L_A - Mg \sin \alpha H = 0$$

$$(+ 4J + MRH) \tau \ddot{\omega}_m + (2N_A + 2N_p) f_v R - 2N_p (L_p + L_A) +$$

$$+ Mg (\cos \alpha L_A - \sin \alpha H) = 0$$

$$\underbrace{2126}_{(+ 4J + MRH) \tau \ddot{\omega}_m} + \underbrace{19277}_{Mg \cos \alpha (f_v R + L_A)} - \underbrace{837}_{Mg \sin \alpha H} = 2N_p (L_p + L_A)$$

$$N_p = 5126 \text{ N}$$



$$\Sigma M_c = 0 \Rightarrow 2J \ddot{\omega}_u + 2T_p R + 2N_p f_v R = 0 \Rightarrow T_p = - \frac{J \tau \ddot{\omega}_m}{R} - N_p f_v$$

$$T_p = - 300 \text{ N}$$

$$T_{lim} = f_s N_p = 4100 \text{ N} \Rightarrow |T_p| < T_{lim} \text{ ADERENZA OKAY}$$

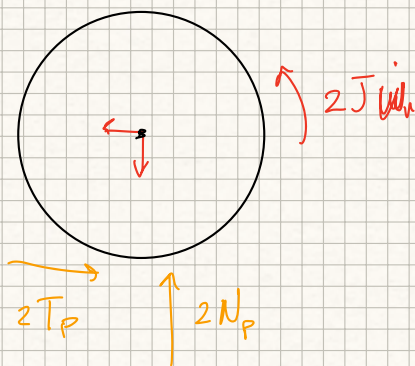
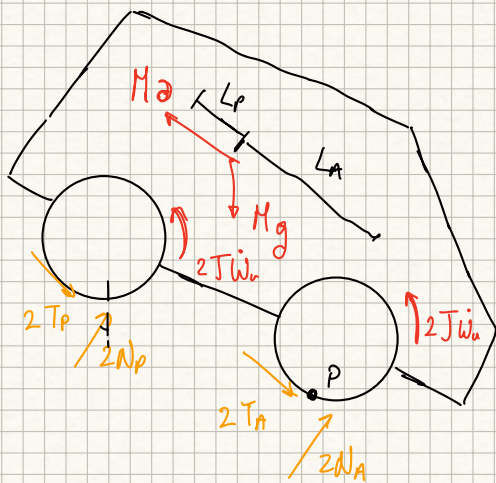
a) $\alpha = 1 \text{ m/s}^2 \Rightarrow \ddot{\omega}_m = 16.67 \text{ rad/s}^2$

$$W_2 = (C_u^* - J_u^* \ddot{\omega}_m) \ddot{\omega}_m = -62.38 \text{ Wm} \Rightarrow \text{MOTO DRETTA}$$

$$J_D C_m - J_D J_m \ddot{\omega}_m + C_u^* - J_u^* \ddot{\omega}_m = 0 \Rightarrow C_m = \frac{J_D J_m \ddot{\omega}_m - C_u^* + J_u^* \ddot{\omega}_m}{J_D} = 81.3 \text{ Nm}$$

La ruota anteriore è in aderenza?

$$N_P = 5126, N_A = 2204$$

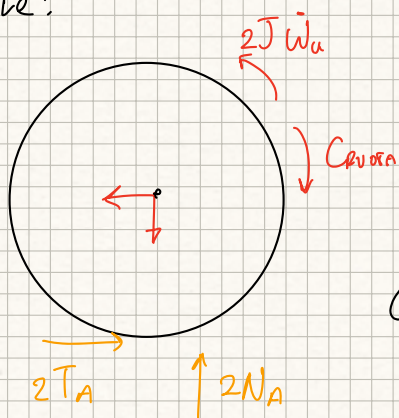


$$2 \cdot T_P R + 2 N_P f_v R + 2 J \tau \dot{\omega}_m = 0$$

$$T_P = -\frac{J \tau \dot{\omega}_m}{R} - N_P f_v = -300 \text{ N}$$

$$2T_P + 2T_A - M\ddot{a} + Mg \sin \alpha = 0 \Rightarrow T_A = 1129 \text{ N} \quad T_{\text{lim}} = f_3 N_A = 1763 \text{ N}$$

Oppure:



$$2T_A R + 2N_A f_v R + 2J \tau \dot{\omega}_m R - C_{\text{gyroTA}} = 0$$

$$T_A + N_A f_v + J \tau \dot{\omega}_m - \frac{C_{\text{gyroTA}}}{2R} = 0$$

$$C_{\text{gyroTA}} = -(C_n^* - J_n^* \dot{\omega}_m) / \tau = 773.8 \text{ Nm}$$

$$T_A = \frac{C_{\text{gyroTA}}}{2R} - N_A f_v - J \tau \dot{\omega}_m = 1165 \text{ N}$$